

Quantitative localization of nonreal zeros of a cosine–Riemann Ξ mixture

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Abstract

For $0 < t < 1$, let $H_t(z) = (1 - t) \cos z + t\Xi(z)$, where $\Xi(z) = \xi(\frac{1}{2} + iz)$ is the Riemann Ξ -function. We construct an explicit sequence of centers, expressed through the principal branch of the Lambert W -function and one elementary correction, such that the disk of radius $40/((2n + 1)\pi)$ about the n th center contains exactly one simple zero of H_t for every sufficiently large n . These disks lie in the first quadrant and are eventually pairwise disjoint. For the resulting zeros $z_n = x_n + iy_n$, we obtain $x_n \sim 4\pi n / \log n$ and $y_n \sim 2\pi^2 n / (\log n)^2$, together with spacing estimates and the relation $y_n = \pi x_n / (2 \log x_n) + O_t(x_n / (\log x_n)^2)$. An explicit sufficient criterion makes the localization applicable at an individual index. The estimates are uniform when t ranges over a fixed compact subset of $(0, 1)$. We also give the qualitative background: the large real zeros follow the cosine lattice, whereas infinitely many nonreal zeros escape every fixed horizontal strip. The localized sequence is not asserted to exhaust all nonreal zeros, and no conclusion about the Riemann hypothesis is drawn.

Keywords. Riemann ξ -function; zeros of entire functions; Lambert W -function; asymptotic inversion; Rouché’s theorem.

1 Introduction and main results

We use the normalization

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right), \quad (1.1)$$

with the removable singularities filled in. The completed function is entire and satisfies $\xi(s) = \xi(1-s)$; consequently Ξ is even and real on the real axis [1, §25.4]. Throughout the paper,

$$H_t(z) = (1 - t) \cos z + t\Xi(z), \quad 0 < t < 1, \quad (1.2)$$

and all zeros are counted with multiplicity unless stated otherwise.

The two terms in (1.2) have different asymptotic roles. In each fixed horizontal strip, $\Xi(z)$ tends to zero as $|\operatorname{Re} z| \rightarrow \infty$, so the large real zeros of H_t follow the cosine lattice. On the imaginary axis, $\log \Xi(iy) = \frac{1}{2} y \log y + O(y)$, which grows faster than the logarithm of $\cosh y$. A classical factorization argument then forces infinitely many nonreal zeros. Appendix A gives a self-contained proof of this qualitative statement and its limitations. The principal result below is an individual quantitative localization of an infinite sequence of nonreal zeros.

1.1 Explicit centers

Set

$$a = 2\pi e^3, \quad c_t = \log\left(2e^{1/2} \sqrt{\pi} \frac{t}{1-t}\right), \quad D_t = 3 \log a + 2c_t - 3. \quad (1.3)$$

For every $q > 0$, define

$$B(q) = W_0\left(-\frac{2iq}{a}\right), \quad u(q) = -\frac{2iq}{B(q)}, \quad h_t(q) = -3 - \frac{D_t}{B(q) + 1}, \quad (1.4)$$

where W_0 is the principal branch of Lambert W . The branch and an alternative real-variable definition are established in Section 4. For $n = 0, 1, 2, \dots$, put

$$\begin{aligned} q_n &= (2n + 1)\pi, \\ \hat{w}(q, t) &= u(q) + h_t(q), \\ \hat{z}_n(t) &= i(\hat{w}(q_n, t) - \tfrac{1}{2}) = i\left(-\frac{2iq_n}{B(q_n)} - \frac{7}{2} - \frac{D_t}{B(q_n) + 1}\right). \end{aligned} \quad (1.5)$$

These centers do not require solving $H_t(z) = 0$.

Theorem 1.1 (Individual localization). *For each fixed $t \in (0, 1)$ there is an integer $N(t)$ such that, for every $n \geq N(t)$, the disk*

$$\mathbb{D}_n(t) = \left\{ z \in \mathbb{C} : |z - \hat{z}_n(t)| < \frac{40}{q_n} \right\} \quad (1.6)$$

lies in the first open quadrant and contains exactly one zero $z_n(t)$ of H_t , counted with multiplicity. In particular, this zero is simple and

$$|z_n(t) - \hat{z}_n(t)| < \frac{40}{(2n + 1)\pi}. \quad (1.7)$$

After increasing $N(t)$, the disks are pairwise disjoint and the sequence $\operatorname{Re} z_n(t)$ is strictly increasing. The threshold can be chosen uniformly for t in any fixed compact subset of $(0, 1)$.

Theorem 1.2 (Geometry and spacing of the localized sequence). *Write $z_n(t) = x_n(t) + iy_n(t)$ and $B_n = B(q_n)$. Then, as $n \rightarrow \infty$,*

$$x_n(t) = \frac{4\pi n}{\log n} \left(1 + O_t\left(\frac{\log \log n}{\log n}\right) \right), \quad (1.8)$$

$$y_n(t) = \frac{2\pi^2 n}{(\log n)^2} \left(1 + O_t\left(\frac{\log \log n}{\log n}\right) \right). \quad (1.9)$$

The centers satisfy

$$\hat{z}_{n+1}(t) - \hat{z}_n(t) = \frac{4\pi}{B_n + 1} + O_t\left(\frac{1}{q_n(\log q_n)^2}\right), \quad |\hat{z}_{n+1}(t) - \hat{z}_n(t)| \sim \frac{4\pi}{\log n}. \quad (1.10)$$

For the zeros themselves,

$$z_{n+1}(t) - z_n(t) = \frac{4\pi}{B_n + 1} + O_t(q_n^{-1}). \quad (1.11)$$

Moreover,

$$y_n(t) = \frac{\pi x_n(t)}{2 \log x_n(t)} + O_t\left(\frac{x_n(t)}{(\log x_n(t))^2}\right). \quad (1.12)$$

All these estimates are uniform on every fixed compact $K \subset (0, 1)$.

The index n in these statements is a phase index for the constructed sequence, not an index in a complete ordering of all nonreal zeros. The ratio of the localization radius to the neighboring-center spacing is $O(\log n/n)$ and tends to zero. Proposition 5.1 gives a sufficient criterion at any individual index; no minimal numerical value of $N(t)$ or optimal value of the constant 40 is asserted.

1.2 Relation to earlier work

Lagarias and Montague [2, Theorem 2.2] study constant levels of the primitive of ξ . Their level curves exhibit a related leading geometry. Their problem is distinct from the equation used here,

$$\xi(w) = -\frac{1-t}{t} \cosh(w - \tfrac{1}{2}),$$

in which the right-hand side is not constant. Csordas and Smith [3, Theorem 1.3] study combinations of real shifts of an entire function. Their discussion of Lindelöf’s criterion on manuscript p. 5 explains the classical growth obstruction underlying our qualitative background; it does not specify individual centers for the additive family (1.2).

The leading inversion by Lambert W , followed by a correction, belongs to the standard asymptotic-inversion method; inverse-gamma examples are discussed by Borwein and Corless [4, pp. 18–19]. Here we keep the complex phase, the zeta and hyperbolic-cosine remainders, and the disk-wide bounds needed to count zeros with multiplicity. The result presented is an explicit localization theorem for (1.2), not a new general inversion principle. The comparison with earlier work is confined to the cited statements.

The proof proceeds by an exact logarithmic reduction in a zero-free right half-plane, explicit estimates for its complex remainder, inversion of the leading phase, and Rouché’s theorem on disjoint disks. The qualitative result in Appendix A is not a premise of this construction. None of the arguments assumes the Riemann hypothesis.

2 Exact logarithmic reduction

Use the change of variable

$$w = \frac{1}{2} - iz = y + \frac{1}{2} - ix, \quad z = i(w - \tfrac{1}{2}), \quad z = x + iy. \quad (2.1)$$

Reflection in (1.1) gives

$$\Xi(z) = \xi(w), \quad \cos z = \cosh(w - \tfrac{1}{2}). \quad (2.2)$$

The map in (2.1) preserves distances and multiplicities.

On the simply connected half-plane $\operatorname{Re} w > 1$, we use $\operatorname{Log} w$ and $\operatorname{Log}(w - 1)$ normalized to be real on $w > 1$, and the holomorphic logarithm of $\Gamma(w/2)$ normalized on the positive real axis. This last function is not the principal logarithm of a numerical gamma value. We also use

$$L_\zeta(w) = \sum_p \sum_{k \geq 1} \frac{p^{-kw}}{k}, \quad \operatorname{Re} w > 1, \quad (2.3)$$

where the sum is over the ordinary primes. The series is absolutely and locally uniformly convergent, and $\exp L_\zeta = \zeta$ by the Euler product [1, §25.2]. Finally, $\log(1 + e^{1-2w})$ denotes its convergent power-series logarithm, since $|e^{1-2w}| < e^{-1}$.

The functions w , $w - 1$, $\Gamma(w/2)$, $\zeta(w)$ and $\cosh(w - \frac{1}{2})$ are nonzero in this half-plane. Define

$$\begin{aligned} L_t(w) = & \log \frac{t}{1-t} + \operatorname{Log} w + \operatorname{Log}(w - 1) - \frac{w}{2} \log \pi + \log \Gamma(w/2) - w + \frac{1}{2} \\ & + L_\zeta(w) - \log(1 + e^{1-2w}). \end{aligned} \quad (2.4)$$

Then exactly

$$e^{L_t(w)} = \frac{t\xi(w)}{(1-t)\cosh(w - \frac{1}{2})}, \quad (2.5)$$

$$H_t(i(w - \tfrac{1}{2})) = (1-t)\cosh(w - \tfrac{1}{2})(1 + e^{L_t(w)}). \quad (2.6)$$

Consequently all zero conditions in this domain are

$$L_t(w) = (2m + 1)\pi i, \quad m \in \mathbb{Z}. \quad (2.7)$$

The sequence in Theorem 1.1 will correspond to

$$L_t(w) = -iq_n, \quad m = -n - 1. \quad (2.8)$$

For a fixed level, multiplicities are preserved because $1 + e^L$ has a simple zero as a function of L at every odd multiple of πi . No logarithm of H_t is being assumed.

Let

$$F(w) = \frac{w}{2} \operatorname{Log}(w/a), \quad F_0(w) = F(w) + \frac{3}{2} \operatorname{Log} w + c_t, \quad (2.9)$$

where the same principal logarithm also defines F on the cut plane. Set

$$S_\Gamma(w) = \log \Gamma(w/2) - \left[\left(\frac{w}{2} - \frac{1}{2} \right) \operatorname{Log}(w/2) - \frac{w}{2} + \frac{1}{2} \log(2\pi) \right]. \quad (2.10)$$

Regrouping (2.4) gives the exact decomposition

$$\begin{aligned} L_t(w) &= F_0(w) + E(w), \\ E(w) &= \operatorname{Log}(1 - 1/w) + S_\Gamma(w) + L_\zeta(w) - \log(1 + e^{1-2w}). \end{aligned} \quad (2.11)$$

The identity $\operatorname{Log}(w - 1) = \operatorname{Log} w + \operatorname{Log}(1 - 1/w)$ follows on $w > 1$ and then throughout the half-plane by analytic continuation. The coefficient of the linear term in (2.9) is $-3w/2$, and the remaining constant is exactly c_t in (1.3).

3 Uniform complex remainder estimates

Lemma 3.1. *For $\sigma = \operatorname{Re} w \geq 2$, the remainder in (2.11) satisfies*

$$|E(w)| \leq \frac{1}{|w| - 1} + \frac{1}{3|w|} + 3 \cdot 2^{-\sigma} + \frac{e^{1-2\sigma}}{1 - e^{1-2\sigma}}, \quad (3.1)$$

and

$$|E'(w)| \leq \frac{10}{|w|^2} + 5 \cdot 2^{-\sigma} + 3e^{1-2\sigma}. \quad (3.2)$$

The estimates are uniform in $\operatorname{Im} w$ and independent of t .

Proof. The power series gives $|\operatorname{Log}(1 - 1/w)| \leq 1/(|w| - 1)$. The logarithmic Stirling expansion and its sector remainder estimate [1, §5.11(i)–(ii)] have first omitted term $1/(6w)$. In the closed right sector the multiplier is $\sec^2(\arg w/2) \leq 2$. Hence

$$|S_\Gamma(w)| \leq \frac{1}{3|w|}.$$

Positivity of the coefficients of (2.3) yields

$$\begin{aligned} |L_\zeta(w)| &\leq \log \zeta(\sigma) \leq \zeta(\sigma) - 1 \\ &\leq 2^{-\sigma} + \int_2^\infty x^{-\sigma} dx \leq 3 \cdot 2^{-\sigma}. \end{aligned} \quad (3.3)$$

The bound $|\log(1 + \eta)| \leq |\eta|/(1 - |\eta|)$ supplies the last term of (3.1).

To estimate the derivative of S_Γ , put $R = |w|$. On the Cauchy circle $|\nu - w| = R/4$, one has $|\nu| \geq 3R/4$ and $|\arg \nu| < 3\pi/4$. The function S_Γ is holomorphic in the cut plane $|\arg \nu| < \pi$, so the same Stirling estimate, now using $\sec^2(3\pi/8) < 7$, gives

$$|S_\Gamma(\nu)| < \frac{2}{R}, \quad |S'_\Gamma(w)| \leq \frac{8}{R^2}.$$

This use of a larger sector concerns only S_Γ ; the Euler logarithm is not continued to the Cauchy circle. Also,

$$\left| \frac{d}{dw} \operatorname{Log}(1 - 1/w) \right| = \frac{1}{|w(w-1)|} \leq \frac{2}{|w|^2}.$$

Termwise differentiation of (2.3) is justified by local uniform absolute convergence. With the von Mangoldt coefficients bounded by $\log n$,

$$|L'_\zeta(w)| \leq \sum_{n \geq 2} (\log n) n^{-\sigma} \leq 4 \cdot 2^{-\sigma} \sum_{n \geq 2} \frac{\log n}{n^2} < 5 \cdot 2^{-\sigma}.$$

Indeed, the summand is decreasing on $[2, \infty)$ and

$$\sum_{n \geq 2} \frac{\log n}{n^2} \leq \frac{\log 2}{4} + \int_2^\infty \frac{\log x}{x^2} dx = \frac{\log 2}{4} + \frac{1 + \log 2}{2} < \frac{5}{4}.$$

Finally, as $e^{1-2\sigma} \leq e^{-3} < 1/3$,

$$\left| \frac{d}{dw} [-\log(1 + e^{1-2w})] \right| \leq \frac{2e^{1-2\sigma}}{1 - e^{1-2\sigma}} < 3e^{1-2\sigma}.$$

Adding these estimates proves (3.2). $\square$

Retaining the first Stirling correction gives, uniformly for $\operatorname{Re} w = \sigma \geq 2$ as $|w| \rightarrow \infty$,

$$E(w) = -\frac{5}{6w} + O(|w|^{-2} + 2^{-\sigma} + e^{-2\sigma}). \quad (3.4)$$

This refinement is not needed for the radius in (1.7).

For clarity, the modulus and phase equations can be written separately. Let $w = \sigma - ix$ with $\sigma > 1$ and $x > 0$, and put $\varrho = |w|$, $\theta = \arctan(x/\sigma)$ and $A = \log(\varrho/a)$. Equation (2.8) is equivalent to the pair

$$\frac{\sigma A - x\theta}{2} + \frac{3}{2} \log \varrho + c_t + \operatorname{Re} E(w) = 0, \quad (3.5)$$

$$\frac{x A + \sigma\theta}{2} + \frac{3}{2} \theta - \operatorname{Im} E(w) = q_n. \quad (3.6)$$

The estimate (3.1) applies when $\sigma \geq 2$. Neither of (3.5) and (3.6) alone replaces the complex equation.

4 The model centers and the Lambert branch

For $q > 0$, let $\beta = \beta(q) \in (0, \pi/2)$ be the solution of

$$e^{\beta \tan \beta} \frac{\beta}{\cos \beta} = \frac{2q}{a}. \quad (4.1)$$

The logarithmic derivative of the left-hand side is

$$\frac{1}{\beta} + 2 \tan \beta + \beta \sec^2 \beta > 0.$$

The left-hand side tends to zero at $\beta = 0$ and to infinity at $\beta = \pi/2$. Thus the solution is unique and smooth in q . Define $\alpha = \beta \tan \beta$ and $B = \alpha - i\beta$. Direct multiplication gives

$$Be^B = -ie^\alpha \frac{\beta}{\cos \beta} = -\frac{2iq}{a}. \quad (4.2)$$

Consequently,

$$u = ae^B = -\frac{2iq}{B}, \quad \operatorname{Re} u > 0, \quad \operatorname{Im} u < 0, \quad \operatorname{Log}(u/a) = B, \quad F(u) = -iq. \quad (4.3)$$

In particular, both the modulus and the phase of the leading equation are satisfied.

As $q \downarrow 0$, $B \rightarrow 0$, and $B + 1 \neq 0$ along this curve. The inverse function theorem continues the local inverse of Be^B at $B = 0$ along the negative imaginary ray. This is the principal branch W_0 , with the standard convention of [1, §4.13]. Hence (4.1) agrees with (1.4) and supplies a real-variable definition independent of any numerical branch selection.

Write

$$U = |u|, \quad s = \operatorname{Re} u, \quad v = -\operatorname{Im} u. \quad (4.4)$$

Then (4.2) implies

$$U = \frac{2q}{|B|}, \quad s = \frac{2q\beta}{\alpha^2 + \beta^2}, \quad v = \frac{2q\alpha}{\alpha^2 + \beta^2}. \quad (4.5)$$

Here s is only the real part of the model center u ; it is not the complex argument in (1.1).

As $q \rightarrow \infty$, $\beta \rightarrow \pi/2$ and $\alpha \rightarrow \infty$. Taking absolute values in (4.2), with $\ell = \log(2q/a)$, gives

$$\begin{aligned} \ell &= \alpha + \frac{1}{2} \log(\alpha^2 + \beta^2), \\ \alpha &= \ell - \log \ell + O\left(\frac{\log \ell}{\ell}\right), \quad \beta = \frac{\pi}{2} + O(\ell^{-1}). \end{aligned} \quad (4.6)$$

Indeed, the first equality initially gives $\alpha \sim \ell$ and then $\alpha = \ell - \log \alpha + O(\alpha^{-2})$. Substitution yields the stated expansion. The estimate for β follows from $\alpha = \beta \cot(\pi/2 - \beta)$. In particular,

$$U \sim \frac{2q}{\log q}, \quad s \sim \frac{\pi q}{(\log q)^2}, \quad v \sim \frac{2q}{\log q}. \quad (4.7)$$

Thus the model centers escape to the right before any use of the smallness of $\zeta(w) - 1$ on a localization contour.

Finally, put $d = F'(u) = (B + 1)/2$. Since $\operatorname{Log} u = \log a + B$, the correction in (1.4) is exactly

$$h_t = -\frac{(3/2) \operatorname{Log} u + c_t}{d} = -3 - \frac{3 \log a + 2c_t - 3}{B + 1}. \quad (4.8)$$

For fixed t , $h_t \rightarrow -3$. This is one explicit linear correction; defining the center does not require a solution of $F_0(w) = -iq$.

5 A finite localization criterion

Proposition 5.1. *Let $q = q_n$ and let B, u, h_t, U, s, v be as in (1.4) and (4.4). Suppose that*

$$U \geq 100, \quad |B| \geq 1, \quad |h_t| \leq 4, \quad q \geq 40, \quad s \geq 7, \quad v \geq 6, \quad (5.1)$$

and

$$3 \cdot 2^{5-s} + \frac{e^{11-2s}}{1 - e^{11-2s}} \leq \frac{1}{U}. \quad (5.2)$$

Then the disk $|w - (u + h_t)| < 40/q$ is contained in $\operatorname{Re} w > 1$, $\operatorname{Im} w < 0$, and contains exactly one zero, counted with multiplicity, of $H_t(i(w - \frac{1}{2}))$. This zero is simple.

Proof. Put $\delta = w - (u + h_t)$ and $\eta = h_t + \delta$. On the full closed disk, $|\delta| \leq 40/q \leq 1$ and $|\eta| \leq 5$. Therefore

$$|w| \geq U - 5, \quad \operatorname{Re} w \geq s - 5 \geq 2, \quad \operatorname{Im} w \leq -v + 5 \leq -1. \quad (5.3)$$

Equations (3.1) and (5.2) give

$$|E(w)| \leq \frac{1}{U-6} + \frac{1}{3(U-5)} + \frac{1}{U} < \frac{3}{U}. \quad (5.4)$$

For example, multiplying the middle expression by U bounds it by $100/94 + 100/285 + 1 < 3$ for every $U \geq 100$.

The segment from u to w obeys the same bounds. Since $F''(w) = 1/(2w)$, Taylor's integral remainder and the integral of the logarithmic derivative yield

$$\begin{aligned} |F(u + \eta) - F(u) - d\eta| &\leq \frac{|\eta|^2}{4(U-5)}, \\ |\operatorname{Log}(u + \eta) - \operatorname{Log} u| &\leq \frac{|\eta|}{U-5}. \end{aligned} \quad (5.5)$$

Using $F(u) = -iq$ and $dh_t + (3/2)\operatorname{Log} u + c_t = 0$, we obtain

$$G(w) := L_t(w) + iq = d\delta + \mathcal{R}(w), \quad |\mathcal{R}(w)| < \frac{55}{4(U-5)} + \frac{3}{U} < \frac{20}{U}. \quad (5.6)$$

The final inequality is valid for every $U \geq 100$, since $55U/[4(U-5)] + 3 \leq 5500/380 + 3 < 20$.

On the circle $|\delta| = 40/q$, the identity $q = U|B|/2$ and $\operatorname{Re} B > 0$ give

$$|d\delta| = \frac{40}{U} \frac{|B+1|}{|B|} > \frac{40}{U} > |\mathcal{R}(w)|. \quad (5.7)$$

Rouché's theorem [1, §1.10(iv)] therefore gives exactly one zero of G , counted with multiplicity. Both functions are holomorphic on a neighborhood of the closed disk by (5.3).

It remains to count the zeros of H_t , rather than just one logarithmic level. On the full disk, (5.6) gives the strict bound

$$|G(w)| < \frac{40}{U} \left(1 + \frac{1}{|B|}\right) + \frac{20}{U} \leq \frac{100}{U} \leq 1. \quad (5.8)$$

Because q is an odd multiple of π , $e^{L_t} = -e^G$. All levels $G = 2k\pi i$ with $k \neq 0$ are excluded by (5.8). The factor $(1 - e^G)/G$, extended by -1 at $G = 0$, is analytic and nonzero for $|G| < 1$. Together with (2.6), this preserves the zero count and multiplicity. The single zero of $H_t(i(w - \frac{1}{2}))$ is therefore simple. The isometry (2.1) sends the disk into the first quadrant. $\square$

For every fixed $t \in (0, 1)$, (4.7) and (4.8) imply all inequalities in (5.1) for all sufficiently large q . Moreover,

$$U2^{5-s} \rightarrow 0, \quad Ue^{11-2s} \rightarrow 0,$$

since $s \asymp q/(\log q)^2$ and $\log U = O(\log q)$. Thus (5.2) holds throughout an eventual tail, not merely on a subsequence. This proves the localization and error-bound parts of Theorem 1.1.

The proposition can also be used at a single index whenever its finite hypotheses are rigorously verified. A floating-point check alone is not an interval certificate. No numerical computation is used in the proof.

On the eventual disks, an additional derivative estimate follows from (3.2), (4.7), and $|w - u| \leq 5$:

$$L'_t(w) = \frac{B+1}{2} + O_t(U^{-1}). \quad (5.9)$$

The exponential terms in (3.2) are eventually smaller than U^{-2} ; the explicit derivative of F_0 varies by $O_t(U^{-1})$ on the disk. This also implies nonvanishing of L'_t there in a sufficiently distant tail, although simplicity in Proposition 5.1 did not require this separate argument.

6 Separation, asymptotics, and parameter uniformity

Differentiating the exact identities $F(u(q)) = -iq$ and (4.2) gives

$$u'(q) = -\frac{2i}{B+1}, \quad B'(q) = \frac{B}{q(B+1)}, \quad h'_t(q) = \frac{D_t B}{q(B+1)^3}. \quad (6.1)$$

In particular,

$$u''(q) = \frac{2iB}{q(B+1)^3} = O\left(\frac{1}{q(\log q)^2}\right), \quad h'_t(q) = O_t\left(\frac{1}{q(\log q)^2}\right).$$

These are derivatives of exact expressions, not differentiated asymptotic error terms. Integration on $[q, q+2\pi]$ yields

$$\widehat{w}(q+2\pi, t) - \widehat{w}(q, t) = -\frac{4\pi i}{B(q)+1} + O_t\left(\frac{1}{q(\log q)^2}\right).$$

Multiplication by i proves (1.10).

For strict ordering, write $\widehat{x}(q, t) = -\operatorname{Im} \widehat{w}(q, t)$. Then

$$\widehat{x}'(q, t) = \frac{2(\alpha+1)}{(\alpha+1)^2 + \beta^2} + O_t\left(\frac{1}{q(\log q)^2}\right) \sim \frac{2}{\log q} > 0. \quad (6.2)$$

Hence consecutive center projections on the real axis have gaps asymptotic to $4\pi/\log q_n$, whereas the sum of their radii is $O(q_n^{-1})$. Eventually each projection lies strictly to the left of the next one. This orders and separates every pair of disks, and also orders the real parts of their zeros. The proof of Theorem 1.1 is complete. Adding two localization errors to (1.10) gives (1.11).

From (4.5), (4.8), and (1.7),

$$x_n = \frac{2q_n\alpha_n}{\alpha_n^2 + \beta_n^2} + O_t(1), \quad y_n = \frac{2q_n\beta_n}{\alpha_n^2 + \beta_n^2} + O_t(1). \quad (6.3)$$

Substitution of (4.6) and $q_n = (2n+1)\pi$ proves (1.8) and (1.9).

To derive (1.12) without an extra log log loss, retain the exact parameter relations. They give

$$\frac{s}{v} = \frac{\beta}{\alpha}, \quad \frac{U}{v} = 1 + O(\alpha^{-2}), \quad \alpha = \log(U/a) = \log v - \log a + O(\alpha^{-2}).$$

Using $\beta = \pi/2 + O(\alpha^{-1})$, it follows that

$$s = \frac{\pi v}{2 \log v} + O\left(\frac{v}{(\log v)^2}\right).$$

The substitutions $x_n = v + O_t(1)$ and $y_n = s + O_t(1)$ now prove (1.12). Both coordinates tend to infinity.

For a fixed compact $K \subset (0, 1)$, D_t is bounded. Consequently the bounds for h_t and h'_t , the geometric errors, and their eventual thresholds can be chosen uniformly for $t \in K$. All remaining estimates are independent of t . This completes Theorem 1.2 and the compact-parameter assertion of Theorem 1.1.

Scope

The construction gives one zero in each specified disk and an infinite sequence of distinct simple zeros. It does not prove that these are all sufficiently large nonreal zeros of H_t . In particular, no complete nonreal-zero count is inferred from the index n . The threshold $N(t)$ is not optimized, and the quantitative estimates are not asserted uniformly up to $t = 0$ or $t = 1$. Neither trajectories in t nor collision events are classified. All conclusions concern the mixture (1.2), not the zero locations of Ξ itself.

A Qualitative background and the real zero tail

This appendix records the qualitative result for the same family. Its growth obstruction is classical: the even order-one version of Lindelöf's criterion relates finite exponential type to a linear bound on the full zero count; see the discussion in [3, p. 5]. We give the needed argument directly through Hadamard factorization. It is independent of the quantitative construction above.

Theorem A.1. *For each fixed $t \in (0, 1)$, the function H_t has infinitely many nonreal zeros, but only finitely many in any fixed horizontal strip $|\operatorname{Im} z| \leq M$. Its sufficiently large real zeros are simple and, on the positive half-axis, can be written*

$$r_n(t) = \left(n + \frac{1}{2}\right) \pi + O_t(n^{-1}). \quad (\text{A.1})$$

Apart from these and their negatives, it has only finitely many real zeros. Thus

$$N_{\mathbb{R}}(R; t) = \frac{2R}{\pi} + O_t(1). \quad (\text{A.2})$$

There are no zeros on the imaginary axis, and there are infinitely many zeros in each open quadrant. In every fixed vertical strip $|\operatorname{Re} z| \leq A$ there are only finitely many zeros.

A.1 The kernel, strip decay, and growth

The classical cosine-transform representation [2, (1.2) and Lemma 3.1] is

$$\Xi(z) = 2 \int_0^\infty \Phi(u) \cos(zu) \, du, \quad (\text{A.3})$$

where

$$\Phi(u) = \sum_{m=1}^\infty \left(4\pi^2 m^4 e^{9u/2} - 6\pi m^2 e^{5u/2}\right) e^{-\pi m^2 e^{2u}}, \quad u \geq 0. \quad (\text{A.4})$$

Each summand is positive, since its prefactor is $2\pi m^2 e^{5u/2}(2\pi m^2 e^{2u} - 3) > 0$. The series and its derivatives converge locally uniformly, and

$$|\Phi(u)| + |\Phi'(u)| \leq C e^{13u/2} \exp\left(-\frac{\pi}{2} e^{2u}\right). \quad (\text{A.5})$$

Indeed, one derivative produces only powers of m of degree at most six and exponentials $e^{ju/2}$ with $j \leq 13$. For $x = e^{2u} \geq 1$,

$$e^{-\pi m^2 x} \leq e^{-\pi x/2} e^{-\pi m^2/2},$$

and the remaining polynomially weighted Gaussian sums converge.

For $z = x + iy$, split (A.3) into its two exponential integrals. If $|y| \leq L$, their amplitudes and first derivatives have common integrable bounds by (A.5). Integration by parts therefore gives

$$|\Xi(x + iy)| \leq \frac{C_L}{|x|}, \quad |y| \leq L, \quad x \neq 0, \quad (\text{A.6})$$

where, for example,

$$C_L = 2\Phi(0) + 2 \int_0^\infty e^{Lu} (|\Phi'(u)| + L\Phi(u)) \, du < \infty.$$

Reflection gives $\Xi(iy) = \xi(\frac{1}{2} + y) > 0$. Stirling's formula and $\zeta(\sigma) = 1 + O(2^{-\sigma})$ as $\sigma \rightarrow +\infty$ imply

$$\log \Xi(iy) = \frac{y}{2} \log \frac{y}{2\pi e} + \frac{7}{4} \log y + \frac{1}{4} \log \frac{\pi}{2} + O(y^{-1}). \quad (\text{A.7})$$

The constants follow directly by substituting $s = y + \frac{1}{2}$ in (1.1); in particular the logarithmic coefficient is $2 - 1/4 = 7/4$. Since the cosine contribution is exponentially smaller,

$$\log H_t(iy) = \log t + \frac{y}{2} \log \frac{y}{2\pi e} + \frac{7}{4} \log y + \frac{1}{4} \log \frac{\pi}{2} + O_t(y^{-1}). \quad (\text{A.8})$$

Positivity in (A.3) gives

$$|\Xi(z)| \leq 2 \int_0^\infty \Phi(u) \cosh(|z|u) du = \Xi(i|z|).$$

Together with $|\cos z| \leq \cosh |z|$, this gives the exact maximum-modulus identity

$$M_{H_t}(r) := \max_{|z|=r} |H_t(z)| = (1-t) \cosh r + t\Xi(ir) = H_t(ir). \quad (\text{A.9})$$

Thus H_t is of order one and infinite exponential type. This conclusion uses no information about its zeros.

A.2 Proof of Theorem A.1

Fix $M > 0$ and $0 < d < \pi/2$. Around the cosine zeros $b_n = (n + \frac{1}{2})\pi$, take disks $D_n = \{z : |z - b_n| < d\}$. On their boundaries,

$$|\cos z| \geq m_d := \min_{|w|=d} |\sin w| > 0.$$

For these circles, apply (A.6) with $L = \max(M, d)$. For all sufficiently large $|n|$, $t|\Xi(z)| < (1-t)m_d$ on ∂D_n . Rouché's theorem gives exactly one zero in each disk, counted with multiplicity. Complex conjugation preserves both H_t and the disk, so this single zero is real and simple.

Outside the disks in the fixed strip, $|\cos z|$ has a positive lower bound: restrict first to a compact fundamental rectangle and then use periodicity. Equation (A.6) therefore excludes all other zeros there when $|\operatorname{Re} z|$ is sufficiently large. The remaining region is compact, and $H_t(0) > 0$ rules out the identically zero function. There are consequently only finitely many nonreal zeros in the strip.

If $r_n = b_n + \delta_n$ is the positive zero, then

$$(1-t)|\sin \delta_n| = t|\Xi(r_n)| \leq \frac{tC_0}{r_n}, \quad |\sin \delta_n| \geq \frac{\sin d}{d} |\delta_n|.$$

This proves (A.1). The fixed spacing of the centers and finitely many central exceptions prove (A.2).

Suppose now that the nonreal zeros are finite in number. Include all exceptional zeros, with multiplicities and symmetries, in a finite even polynomial Q . Hadamard factorization for an entire function of order at most one [5, Theorem XI.3.4] gives

$$H_t(z) = e^{A+Bz} Q(z) \prod_{j \geq 1} \left(1 - \frac{z^2}{\lambda_j^2}\right), \quad (\text{A.10})$$

where λ_j are the remaining positive real zeros and $\lambda_j \geq cj$ for some $c > 0$. The genus-one factors can be paired because $\sum_j \lambda_j^{-2} < \infty$, and $E_1(z/\lambda)E_1(-z/\lambda) = 1 - z^2/\lambda^2$. Evenness forces $B = 0$. For sufficiently large y ,

$$\begin{aligned} \log |H_t(iy)| &\leq C + (\deg Q) \log(1+y) + \sum_{j \geq 1} \log \left(1 + \frac{y^2}{c^2 j^2}\right) \\ &\leq C + (\deg Q) \log(1+y) + \frac{\pi y}{c}. \end{aligned}$$

The last step follows from monotonicity and the elementary integral

$$\int_0^\infty \log(1 + A^2/x^2) dx = \pi A, \quad A > 0.$$

This one-sided linear upper bound contradicts (A.8). Thus there are infinitely many nonreal zeros, and only finitely many of them in every fixed horizontal strip.

For real y , both terms in

$$H_t(iy) = (1 - t) \cosh y + 2t \int_0^\infty \Phi(u) \cosh(yu) du$$

are positive. There are no imaginary-axis zeros. Evenness and conjugation place each nonreal zero in a four-element orbit, one point in each open quadrant; hence every quadrant contains infinitely many zeros.

Finally, for bounded $|x| \leq A$ and $y \rightarrow +\infty$, reflection and uniform right-half-plane Stirling estimates give

$$\log |\Xi(x + iy)| = \frac{1}{2}y \log y + O_A(y), \quad |\cos(x + iy)| \leq e^y.$$

The first summand after multiplication by t eventually dominates the cosine term in modulus. The reverse triangle inequality excludes zeros on that end of the vertical strip. Conjugation handles $y \rightarrow -\infty$, and the remaining rectangle contains only finitely many zeros. $\square$

Proposition A.2 (Qualitative limit as $t \downarrow 0$). *For each fixed $M \geq 0$ there is $\delta_M > 0$ such that, whenever $0 < t < \delta_M$, all nonreal zeros of H_t lie outside $|\operatorname{Im} z| \leq M$.*

Proof. The case $M = 0$ is immediate. Fix $M > 0$, $0 < t_0 < 1$ and $0 < d < \pi/2$. In the tail argument above, bound $t/(1-t)$ by $t_0/(1-t_0)$. A single threshold X then makes every zero with $|\operatorname{Re} z| > X$, $|\operatorname{Im} z| \leq M$ real and simple, for all $0 \leq t \leq t_0$. Enlarge X to an integer multiple of π between successive disks. In the remaining compact rectangle, take finitely many small conjugation-invariant disks about its cosine zeros, with radii less than M . On their boundaries and the compact complement, $|\cos z|$ has a positive minimum. Compact uniform convergence $H_t \rightarrow \cos$ as $t \downarrow 0$ then gives one simple real zero per disk by Rouché and no others on the complement. Combining this with the uniform tail proves the proposition. $\square$

This is a statement about zero sets, not a classification of individual zero trajectories. At the other endpoint, the tail estimates contain $t/(1-t)$ and are not uniform as $t \uparrow 1$; no conclusion about the zeros of Ξ follows by substituting $t = 1$.

B A general growth obstruction

For completeness, the qualitative argument separates from the special arithmetic function as follows. This proposition is a consequence of classical entire-function theory, not an asserted new general principle.

Proposition B.1. *Let f be an even entire function of order at most one, real on the real axis. Suppose that for every $M > 0$,*

$$\sup_{|y| \leq M} |f(x + iy)| \rightarrow 0 \quad (|x| \rightarrow \infty), \quad \limsup_{y \rightarrow \infty} \frac{\log |f(iy)|}{y} = +\infty.$$

Then for every real $a_0 \neq 0$ and $\omega > 0$, the function $G(z) = a_0 \cos(\omega z) + f(z)$ has infinitely many nonreal zeros, only finitely many of them in any fixed horizontal strip, and

$$N_{\mathbb{R}}(R; G) = \frac{2\omega R}{\pi} + O(1).$$

Proof. Choose disks of radius $d < \pi/(2\omega)$ about the cosine lattice $(n + \frac{1}{2})\pi/\omega$, and use the strip bound at height $\max(M, d)$. Rouché and conjugation give a simple real zero in every sufficiently distant disk, with a bounded displacement from its center. A positive lower bound for the cosine on the complement excludes the remaining distant strip zeros. This proves the real count and strip finiteness.

The function G has order at most one. Choose $y_k \rightarrow \infty$ so that $|f(iy_k)| \geq e^{(k+\omega+1)y_k}$. Then eventually $|G(iy_k)| \geq |f(iy_k)|/2$. If the nonreal zeros of G were finite in number, the factorization argument in (A.10), including any even-order zero at the origin in Q , would give $\log |G(iy)| \leq C(1+y)$ for large y . This contradicts the chosen sequence. No rate of the bounded displacement, no positivity of $f(iy)$, and no full limit in place of the limsup are required. $\square$

Corollary B.2. *For every $a_0, b_0, \omega > 0$, the function $a_0 \cos(\omega z) + b_0 \Xi(z)$ satisfies the conclusions of Proposition B.1.*

Proof. The required hypotheses for $f = b_0 \Xi$ follow from (A.6), (A.7), and the maximum-modulus estimate preceding (A.9). $\square$

Statement on AI assistance

AI-based tools were used in problem formulation, development and drafting of the arguments, translation, and separate-context preliminary proof review. The analytical arguments are presented in full. No numerical root calculation or automated review verdict is a premise of a proof. Such assistance and review do not constitute formal verification or external journal peer review.

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